

Inequalities

Problem 1. Let n be a positive integer, and let $x_1, x_2, \dots, x_n$ be positive real numbers such that $x_j \geq 1$ for any j . Prove that

$$1 + x_1 x_2 \cdots x_n \geq \frac{(1 + x_1)(1 + x_2) \cdots (1 + x_n)}{2^{n-1}}.$$

Problem 2. Prove that if $x > 0$ and $n \in \mathbb{N}$, then we have

$$\frac{x^n(x^{n+1} + 1)}{x^n + 1} \leq \left(\frac{x + 1}{2}\right)^{2n+1}.$$

Problem 3. Let $x_1, x_2, \dots, x_n$ be positive real numbers. Prove the inequality

$$\frac{1}{x_1 + 1} + \frac{1}{x_2 + 1} + \cdots + \frac{1}{x_n + 1} \leq \frac{n}{1 + \frac{n}{\frac{1}{x_1} + \frac{1}{x_2} + \cdots + \frac{1}{x_n}}}.$$

Problem 4. Let n be a positive integer, and let $x_1, x_2, \dots, x_n$ be positive real numbers satisfying $x_1 x_2 \cdots x_n = 1$. Prove that there exists $k \in \{1, 2, \dots, n\}$ such that

$$\frac{\sqrt[k]{x_k} + \sqrt[k+1]{x_{k+1}} + \cdots + \sqrt[n]{x_n}}{n + 1 - k} \geq 1.$$

Problem 5. Let a, b, c, d be positive real numbers satisfying $a + b + c + d = 4$. Prove that

$$\frac{a}{b(b+1)} + \frac{b}{c(c+1)} + \frac{c}{d(d+1)} + \frac{d}{a(a+1)} \geq \frac{8}{(a+c)(b+d)}.$$

Problem 6. Prove that for numbers a, b, c in the interval $[0, 1]$,

$$\frac{a}{b+c+1} + \frac{b}{c+a+1} + \frac{c}{a+b+1} + (1-a)(1-b)(1-c) \leq 1.$$

Problem 7. Let $n \geq 3$ be a natural number, and let $a_1, a_2, \dots, a_n, b_1, b_2, \dots, b_n$ be positive real numbers such that $a_1 + a_2 + \cdots + a_n = 1$ and $b_1^2 + b_2^2 + \cdots + b_n^2 = 1$. Prove that

$$a_1(b_1 + a_2) + a_2(b_2 + a_3) + \cdots + a_n(b_n + a_1) < 1.$$

Problem 8. Let $a_1, a_2, a_3, \dots, a_{2011}$ be nonnegative reals with sum $\frac{2011}{2}$. Prove that

$$|(a_1 - a_2)(a_2 - a_3) \cdots (a_{2011} - a_1)| \leq \frac{3\sqrt{3}}{16}.$$